

BEFORE TUESDAY

What to Focus On, Chapters 3 and 8

A narrowed reading list, so the dense parts do not have to slow you down

Both chapters are long, and not every section carries the same weight for Tuesday. Below, each chapter is split into three groups: read closely, skim for the general idea, and skip for now. Tuesday's class explores several of these ideas directly on a live Linux system, so the reading is organized around what that session touches most.

How to use this list. Skip for now does not mean permanently skip, it means these sections are lower priority for this particular class session. This list narrows where to spend your first pass through a long chapter, it does not replace the assigned reading.

1. Chapter 3, A Top-Level View of Computer Function and Interconnection

Read closely

- Computer Components, the von Neumann architecture. This is the vocabulary the rest of the chapter builds on.
- The instruction cycle, fetch cycle, and the four action categories. This is the base cycle that interrupts get added onto.
- Interrupts, Table 3.1 through the discussion of multiple and nested interrupts, Figures 3.9 to 3.14. This is the section Tuesday's class explores directly on a live machine.
- I/O Function. A short section, but it explains why an I/O module exists at all, and it sets up Chapter 8.
- Bus Interconnection, the data bus, address bus, and control bus. Tuesday looks at the physical version of this.

Skim for the idea

- Point-to-Point Interconnect and Quick Path Interconnect. The one idea worth keeping is that a shared bus was replaced by dedicated point to point links, not the layer by layer protocol detail.
- Peripheral Component Interconnect (PCI). You need the general idea of what a PCIe bus does, not the electrical detail.

Skip for now

- QPI Link Layer, QPI Routing and Protocol Layers.
- The PCIe transaction layer packet structure, Table 3.2, and Figure 3.25. This is wire level packet formatting and is not needed for Tuesday.

2. Chapter 8, Input/Output

Read closely

- The generic model of an I/O module, Figure 8.1, and the I/O Functions section. This is the concept behind everything else in the chapter.
- Programmed I/O, I/O Commands, I/O Instructions, and memory-mapped versus isolated I/O, Table 8.1 and Figures 8.4 and 8.5.
- Interrupt-Driven I/O, Figures 8.6 and 8.7. This connects straight back to the interrupt material in Chapter 3.
- Drawbacks of Programmed and Interrupt-Driven I/O. This short section is the argument Tuesday's class watches happen on a live system.
- DMA, Figures 8.12 and 8.13, the general concept of a DMA controller taking over a transfer.
- Universal Serial Bus (USB), and PCI Express and SATA. These are the two external interfaces Tuesday looks at directly.

Skim for the idea

- The Keyboard/Monitor example.
- Device Identification and the general idea of an interrupt controller. You do not need the specific register layout of the 82C59A.
- Alternative DMA Configurations and the Fly-By DMA Controller. You need the idea that a DMA controller sits on the bus, not the signal level detail.
- Evolution of the I/O Function and I/O Channel Architecture, Figure 8.18.
- FireWire, SCSI, Thunderbolt, InfiniBand, Ethernet, and Wi-Fi. Enough to recognize the category, we are not demonstrating all of them.

Skip for now

- Figures 8.8 to 8.11, the Intel 8259A and 8255A register and control word detail.
- Table 8.2, Intel 8237A DMA registers.
- Direct Cache Access, the Xeon E5 chip architecture, and the DDIO comparison, Figures 8.16 and 8.17.
- The IBM z13 I/O channel structure and I/O system structure, Figures 8.19 and 8.20.